LabNo:5 Date: 07/02/2026

INTRODUCTION TO FLEX

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<u>CSE B2 - 37</u>

Lab Exercises:

Q1) Write a FLEX program to count the number of vowels and consonants in the given input.

Ans)

```
%{
#include <stdio.h>

int vowels = 0;
int consonants = 0;
%}

%%
[aeiouAEIOU]        { vowels++; }
[b-df-hj-np-tv-zB-DF-HJ-NP-TV-Z] { consonants++; }
\n                  { /* ignore newline */ }
.                   { /* ignore other characters */ }
%%

int main()
{
   printf("Enter input text:\n");
   yylex();
   printf("Number of vowels     : %d\n", vowels);
   printf("Number of consonants : %d\n", consonants);
   return 0;
}

int yywrap()
{
   return 1;
}
```

Output:

```
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ flex q1.l
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ gcc lex.yy.c -o q1
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ ./q1
Enter input text:
Hello My Name is Satyam Singh
Number of vowels     : 8
Number of consonants : 16
```

Q2)Write a FLEX program to count the number of words, characters, blanks and lines in a given text.

Ans)

```
%{
#include <stdio.h>
#include <string.h>

int words = 0;
int chars = 0;
int blanks = 0;
int lines = 0;
%}

%%
[ \t]+        { blanks += yyleng; chars += yyleng; }
\n            { lines++; chars++; }
[a-zA-Z0-9]+    { words++; chars += yyleng; }
.             { chars++; }
%%

int main()
{
    printf("Enter the text:\n");
    yylex();
    printf("\nNumber of lines      : %d", lines);
    printf("\nNumber of words      : %d", words);
    printf("\nNumber of blanks     : %d", blanks);
    printf("\nNumber of characters : %d\n", chars);
    return 0;
}

int yywrap()
{
    return 1;
}
```

Output:

```
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ flex q2.l
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ gcc lex.yy.c -o q2
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ ./q2
Enter the text:
Hello
My name is
Satyam Singh

Number of lines      : 3
Number of words      : 6
Number of blanks     : 5
Number of characters : 32
```

Q3)Write a FLEX program to find the number of positive integer, negative integer, positive floating positive number and negative
floating point number
Ans)

```
%{
#include <stdio.h>

int posInt = 0;
int negInt = 0;
int posFloat = 0;
int negFloat = 0;
%}

%%
[-][0-9]+"."[0-9]+    { negFloat++; }
[+][0-9]+"."[0-9]+    { posFloat++; }
[0-9]+"."[0-9]+       { posFloat++; }

[-][0-9]+             { negInt++; }
[+][0-9]+             { posInt++; }
[0-9]+                { posInt++; }

[ \t\n]+              ;   /* ignore spaces */
.                     ;   /* ignore other characters */
%%

int main()
{
    printf("Enter input:\n");
    yylex();
    printf("\nPositive Integers        : %d", posInt);
    printf("\nNegative Integers        : %d", negInt);
    printf("\nPositive Floating Numbers: %d", posFloat);
```

```
    printf("\nNegative Floating Numbers: %d\n", negFloat);
    return 0;
}

int yywrap()
{
    return 1;
}
```

Output:

```
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ flex q3.l
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ gcc lex.yy.c -o q3
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ ./q3
Enter input:
2
-8
9.8
-4.7

Positive Integers        : 1
Negative Integers        : 1
Positive Floating Numbers: 1
Negative Floating Numbers: 1
```

Q4)Write a FLEX program to given a input C file, replace all scanf with READ and printf with WRITE statements also find the number
of scanf and printf in the file.

Ans)

```
%{
#include <stdio.h>

int scanfCount = 0;
int printfCount = 0;
%}

%%
scanf          { printf("READ\n"); scanfCount++; }
printf         { printf("WRITE\n"); printfCount++; }
.|\n           { printf("%s", yytext); }   /* print everything else as-is */
%%

int main()
{
    yylex();
    printf("\n\nNumber of scanf  : %d\n", scanfCount);
    printf("Number of printf : %d\n", printfCount);
```

```
    return 0;
}

int yywrap()
{
    return 1;
}
```

Input File:

```
week5 > C input4.c > ...
#include <stdio.h>

int main() {
    int a, b;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    printf("Sum = %d\n", a + b);
    return 0;
}
```

Running commands:

```
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ flex q4.l
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ gcc lex.yy.c -o q4
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ ./q4 < input4.c > output4.c
```

```
week5 > C output4.c > ...
#include <stdio.h>

int main() {
    int a, b;
    WRITE
("Enter two numbers: ");
    READ
("%d %d", &a, &b);
    WRITE
("Sum = %d\n", a + b);
    return 0;
}


Number of scanf  : 1
Number of printf : 2

```

Output File:

Q5)Write a FLEX program to that changes a number from decimal to hexadecimal notation.
Ans)

```
%{
#include <stdio.h>
#include <stdlib.h>
%}

%%
[0-9]+    {
            int num = atoi(yytext);        // Convert string to integer
            printf("Decimal: %d -> Hexadecimal: 0x%X\n", num, num);
          }
.|\n      { printf("%s", yytext); }      // Print other characters as-is
%%

int main() {
   printf("Enter text with decimal numbers:\n");
   yylex();
   return 0;
}

int yywrap() {
   return 1;
}
```

Output:

```
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ flex q5.l
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ gcc lex.yy.c -o q5
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ ./q5
Enter text with decimal numbers:
10 20 30 40
Decimal: 10 -> Hexadecimal: 0xA
 Decimal: 20 -> Hexadecimal: 0x14
 Decimal: 30 -> Hexadecimal: 0x1E
 Decimal: 40 -> Hexadecimal: 0x28
```

Q6)Write a FLEX program to convert uppercase characters to lowercase characters of C file excluding the characters present in the
Comment.
Ans)
```
%{
#include <ctype.h>
#include <stdio.h>
int inComment = 0;  // Flag to track if we are inside a comment
%}

%%
"/*"            { inComment = 1; printf("/*"); }
"*/"            { inComment = 0; printf("*/"); }
"//".*          { printf("%s\n", yytext); }   // Print single-line comments as-is
.               {
                    if (!inComment) {
                        if (isupper(yytext[0]))
                            printf("%c", tolower(yytext[0]));
                        else
                            printf("%c", yytext[0]);
                    } else {
                        printf("%c", yytext[0]); // Inside block comment, print as-is
                    }
                }
\n              { printf("\n"); }
%%

int main() {
    yylex();
    return 0;
}

int yywrap() {
    return 1;
}
```

Input File:

```
week5 > C readfile.c > main()
/*the is a code to add TWO numbers*/
#include<stdio.h>

int main(){
    int a= 6, b = 8;
    printf("THE SUM IS %d", a + b);
    return 0;
}
```

Running Commands:

```
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ flex q6.l
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ gcc lex.yy.c -o q6
cdl-6cse-b2@sce-cl11-03:~/Documents/230905256/week5$ ./q6 <readfile.c > out6.c
```

Output File:

```
week5 > C out6.c > ...
/*the is a code to add TWO numbers*/
#include<stdio.h>

int main(){
    int a= 6, b = 8;
    printf("the sum is %d", a + b);
    return 0;
}
```